

GAME ASSISTANT

Group Y

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MOTIVATION

Games can be:

- Complex
- Hard to learn
- Difficult

Players might:

- Get stuck
- Don’t know / Forget Mechanics
- Feel overwhelmed



Solution:

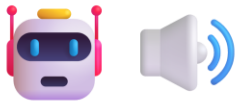
- On-demand support
- Real-time answers
- Accurate information



GOAL

Create an AI voice-to-voice assistant that:

- Listens to questions when they arise
- Answers concise and reliably
- Doesn’t disrupt gameplay



The target scope is: Food in Minecraft

ROLES

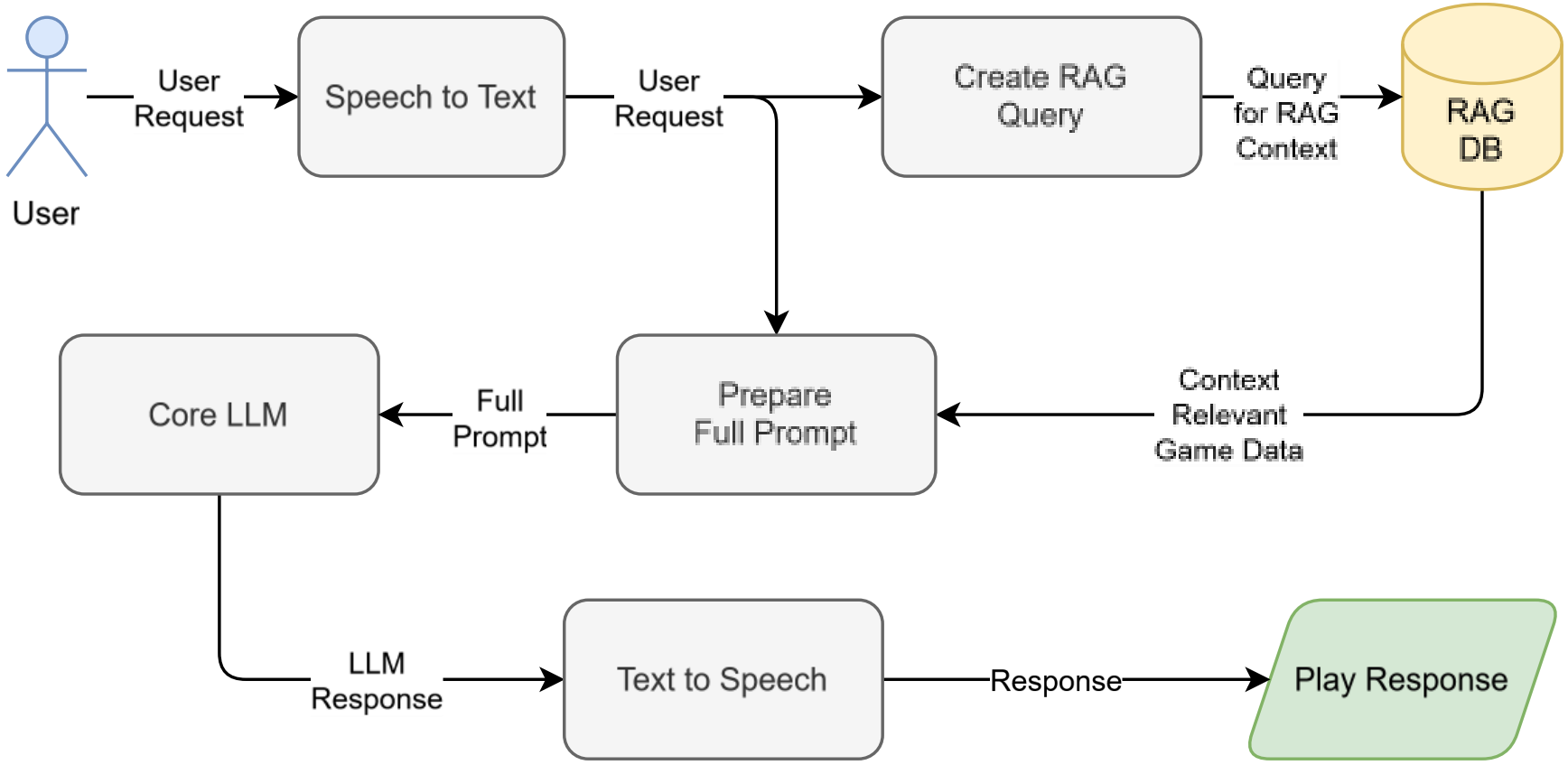
User 

- Asks game related questions
- Listens to the response

Assistant 

- Listens to user input
- Searches DB for information (context)
- Formulates a concise answer
- Answers verbally

SYSTEM STRUCTURE I



SYSTEM STRUCTURE II

LLM

OLLAMA

qwen3:4b-instruct

Multi instance use

RAG

ChromaDB

Multi (2) Collection Layout

Intelligent* Querying

Text-To-Speech

pyttsx3

Speech-To-Text

OpenAI Whisper

CHANGES TO THE IDEA

Planned

Take game state into account

Evaluate responses before output

No GUI

Scope: Food in Minecraft

RAG holds Recipes

Implemented

Game state is **not** observed

LLM handles fault input and missing context

Simple GUI

Scope: Food and Farming in Minecraft

RAG holds Recipes and Guides

IMPLEMENTED TECHNIQUES

RAG Improvements

- Transforming recipes to clear-text
- Converting guides to markdown – Chunking by chapters
- User query intent classification – Guide vs. Recipe
- User query keyword expansion

LLM Improvements

- Start with system prompt to inform about role as Minecraft assistant
- Pass to TTS after every sentence (don't wait for full answer)
- Remove RAG context of previous request to save tokens

OUTCOME

LLM

Works good for the most part.

Some hallucinations.

Sometimes context is interpreted wrong.

RAG

Good performance, good accuracy.

Large context window for guides.

Query processing could be improved.

Text-To-Speech

Uses default system voice in English

Easily understandable

Speech-To-Text

Effectively converts verbal user requests into text for the RAG pipeline.



DEMO

NEXT STEPS

EXPAND SCOPE

Cover all the contents of Minecraft in the assistant.

Collect and preprocess the data (if necessary).

IMPROVE RAG

Switch to Graph RAG to more accurately structure the data of the game.

Add improved query handling to more efficiently search for referenced entities.

IMPROVE GUI AND EXPLAINABILITY ADD ORCHESTRATOR

The GUI does not support switching I/O devices.

Referenced sources should also be evident from a user perspective.

FINE TUNE LLM

..to more easily detect faulty input or faults inside the assistant pipeline; i.e. wrong context.

Prevent hallucination and word repetition.

RESOURCES

- How to Enhance the Performance of Your RAG Pipeline
- Practical Tips and Tricks for Developers Building RAG Applications
- What is LLM Orchestration?
- Microsoft GraphRAG